The Babel Effect: Cross-Lingual Causal Profiles of Factual Recall in Two Small Language Models

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Abstract

Do language models store a fact once, in a language-neutral form, or separately per language? We trace the causal effect of restoring clean activations when the subject of a factual prompt is corrupted, in English (EN), Spanish (ES) and Hindi (HI), for Llama-3.2-3B-Instruct (Grattafiori et al., 2024; 92 facts) and Gemma-2-2B (58 facts), and compare per-layer profiles with a shuffled-fact permutation null. English and Spanish profiles of the same fact are aligned beyond chance in both models (MLP-profile cosine z = 4.22 and 4.81, $p \leq 0.001$; 0.0001 with 10,000 permutations). Hindi overlap is weaker: with a pooled null, Llama's EN-HI and ES-HI pairs are above chance (z = 2.45, 2.47) and Gemma's EN-HI marginally so (z = 2.19), but with a null restricted to the same relation only EN-ES remains significant after correction in both models. The weakness depends partly on Hindi subject length: with short Hindi subject spans the EN-HI cosine is 0.60 (Llama) against 0.20 with long spans. The layer where the fact is looked up is model-dependent (median layer 3 in Llama, 6 in Gemma). Edit transfer over every layer depends on the model: in Gemma mid-layer edits transfer more to Spanish and Hindi than early-layer edits, in Llama the opposite holds for Spanish, and Hindi transfer is small in both. The alignment is robust to rewording (within-language profile cosine exceeds the cross-language cosine in both models) and, in Llama, holds on 104 facts whose answers are not copyable across languages (EN-ES z = 4.87); an English statement in the prompt changes Spanish and Hindi answers in Llama more often than the weight edit does. A residual-stream interchange test then swaps the subject-token state between languages (Section 4.7): the answer follows the swapped state nearly as much as for a same-language swap for English-Spanish (ratio 0.90-1.17 in both models; 95% lower bounds ≥ 0.83 under four bootstrap clusterings) and, when the whole Hindi subject span is patched, for English-Hindi in both directions (ratio 0.81-0.93; every lower bound ≥ 0.70 under four clusterings; the single-token Hindi-to-English estimate of 0.74-0.80 was a lower bound). Controls added after review show that the English-Spanish alignment survives centring of the profiles and entity clustering in both models, but in Llama it is stronger in facts whose English and Spanish subject names share subword pieces (increment 0.40 against 0.30) and still present without them (Section 4.6). A Llama-3.2-3B base replicate keeps all four interchange labels within the registered margin of 0.15 of the instruct model (English to Hindi is the weakest, 0.72, PARTIAL). Pooled over 40 Llama facts, a rank-1 edit at layer 14 reaches Spanish with a median ratio of 0.75 against 0.58 at the traced layer and Hindi with 0.10 against 0.00, at about four times the side-effect cost (Section 4.5). Profile similarity is consistent with partly shared lookup for English and Spanish and cannot be separated from subject-name overlap in Llama; the interchange test supports shared retrieval of the fact at the subject token across all three languages in both models, on four geography-type relations and two small models, and does not by itself show a single language-agnostic space.

1 Introduction

Multilingual language models answer factual questions in many languages, yet it is unclear whether a fact is stored once and read out through language-specific front and back ends, or stored separately per language. The answer determines whether an edit made in one language should carry over to others. We test it with

two measurements on open-weight models that fit one 8 GB GPU: overlap of causal-tracing profiles across languages, and transfer of a localised edit from English to Spanish and Hindi.

Contributions: (i) a cross-lingual causal-tracing comparison with a shuffled-fact permutation null, a relation-stratified variant of it and a Hindi tokenisation diagnostic; (ii) results on two models with different lookup layers; (iii) an edit-transfer measurement over every layer, with robustness checks (rewording, non-copyable relations, an in-context baseline); (iv) a pre-registered design with logged deviations, all code, write-once run directories and a verifiable claim manifest.

2 Related work

Locating and editing facts. Causal tracing and rank-one editing of MLP weights follow Meng et al. (2022, ROME, arXiv:2202.05262) and its multi-edit extension MEMIT (Meng et al., 2022, arXiv:2210.07229). Geva et al. (2023, arXiv:2304.14767) describe recall as subject enrichment, relation propagation and attribute extraction. Hase et al. (2023, arXiv:2301.04213) find that the layer where tracing peaks does not reliably predict the best layer to edit, which motivates sweeping edit layers rather than relying on the traced layer. For activation patching itself we follow the advice of Zhang and Nanda (2023, arXiv:2309.16042) and Heimersheim and Nanda (2024, arXiv:2404.15255) on metrics and corruption choices, and we use their point that results depend on these choices to report both a same-language ceiling and a lexical-overlap control. Gupta et al. (2024, arXiv:2403.07175) and Yang et al. (2024, arXiv:2406.11263) trace model collapse after ROME edits to implementation details, which is relevant to how damage is measured.

Multilingual recall mechanisms. Wendler et al. (2024, arXiv:2402.10588) report that Llama-2 middle layers decode towards English before language-specific output. Fierro et al. (2024, arXiv:2410.14387) use patching on XGLM, EuroLLM and mT5 and find subject enrichment to be language-independent and object extraction language-dependent. Zhang et al. (2024, arXiv:2406.16416) locate language-agnostic factual neurons mostly in middle layers. Wang et al. (2025, arXiv:2504.04264) study cross-lingual factual inconsistency in LLaMA2-7B and BLOOM-560M. Lu et al. (2025, arXiv:2505.20546) analyse the multilingual factual-recall pipeline of Llama-3.2-3B, the model used here, in English, French, Spanish, Chinese, Japanese and Korean, without editing and without Hindi. Schut et al. (2025, arXiv:2502.15603) report English-like internal decisions in multilingual models. Dumas et al. (2024, arXiv:2411.08745) patch latents from a source translation prompt into a target prompt and find that the output language is encoded earlier than the concept, so the concept can be changed without changing the language; their task is word translation, whereas our interchange test (Section 4.7) swaps the subject of a factual-recall prompt across languages. Multilingual factual probes are provided by X-FACTR (arXiv:2010.06189) and mLAMA (arXiv:2102.00894).

Cross-lingual knowledge editing. Wang et al. (2023, arXiv:2309.08952) and Khandelwal et al. (2024, arXiv:2407.10275) find that parameter-based editing transfers poorly across languages while in-context editing transfers much better; Nie et al. (2024, arXiv:2406.17764) report in-context editing across 53 languages and a penalty for non-Latin scripts. Qi et al. (2023, arXiv:2310.10378) edit BLOOM-3b with ROME in English and find that the edit propagates to languages whose factual knowledge is more consistent with English, including Spanish. Banerjee et al. (2024, arXiv:2406.11139) apply ROME and MEMIT in English, Spanish, Hindi and five other languages on 7B models and report low portability across languages. Beniwal et al. (2024, arXiv:2401.10521) study cross-lingual editing in BLOOM, mBERT and XLM-RoBERTa, including Hindi, and find the largest limitations when the two languages use different scripts. Mousi et al. (2025, arXiv:2507.09629) find on Llama-2-7B-chat that parameter-based editing (ROME, MEMIT) generalises poorly from English to Arabic. Ifergan et al. (2024, arXiv:2408.10646) use editing itself as a probe of shared representation and report that high cross-lingual answer consistency does not imply shared representation, with script similarity the dominant factor; their result and ours agree that script matters, but we measure sharing with an interchange test at the subject token and relate it to edit transfer on the same model. Durrani

et al. (2025, arXiv:2505.14393) survey the area.

Position of this work. The studies above either trace or edit, mostly on 7B-class models, and none of them compares per-layer causal profiles across English, Spanish and Hindi against a shuffled-fact null and then measures edit transfer, specificity and damage over every layer on the same model. We searched for such a combination and did not find one; this is an absence in a bounded search, not proof that none exists.

3 Method

Models. Llama-3.2-3B-Instruct (28 layers) and Gemma-2-2B (26 layers, base model), plus Llama-3.2-3B base (same architecture and tokenizer as the instruct model) for the interchange replication of Section 4.7 only, bf16, TransformerLens 3.9.0 without weight processing; one 8 GB GPU (peak 7.5 GiB). Weights are byte-identical (SHA-256) to the official repositories. Gemma is a base model and Llama an Instruct model; this confound is not separable here. A third family (BLOOM-1b7, pre-registered as A10, to test whether the weak Hindi overlap is a tokenisation effect) had not been run when this draft was written.

Facts and phrasings. Hand-written EN/ES/HI facts in five relations (capital_of, capital_country, landmark_city, invention_inventor, event_year; 408 candidates, 447 for Gemma after adding 39 events), three phrasings per fact and language. For each relation and language the three best of six answer-first templates are chosen on a 30-fact tuning set (per model). A fact is kept if the canonical phrasing begins with the answer (strict, after spelling normalisation) in all three languages and one other phrasing contains it in its first line. Kept: 93 (Llama: capital_of 27, capital_country 38, landmark_city 6, invention_inventor 0, event_year 22) and 58 (Gemma: 11, 11, 6, 0, 30).

Causal tracing. Gaussian noise (sigma = 3 $\times$ pooled subject-embedding std, per-fact seed) is added to the subject-token embeddings. A clean activation is restored at one (layer, prompt token) site of the residual stream or MLP output, and the effect is the change in the teacher-forced log-probability of the whole answer relative to the corrupted run (full-answer scoring; the first-token variant is stored). A pair is valid if the answer probability falls below half its clean value and the total effect is positive (92 of 93 and 58 of 58 facts).

Overlap. Per fact and language pair, cosine and Spearman correlation of the 28 (or 26) layer MLP-output profile at each language's own last subject token; statistic = median over facts; null = fact i in one language paired with a permuted fact in the other (1000 permutations; pooled across relations, and, as a post-hoc sensitivity analysis that becomes primary for later data, within relation); z and one-sided p from the null; 95% CIs by 2000 bootstrap resamples of facts. Six primary tests per model (3 pairs x 2 statistics), Bonferroni alpha 0.0083.

Editing. Rank-1 update of the MLP down-projection (ROME-style) at every layer for 10 facts per model, English prompt only, fitted on the full new answer; transfer ratio, specificity (20 controls) and damage (perplexity ratio on 100 WikiText-2 test sentences; Merity et al., 2016) as in A5. An in-context baseline (A9) is run for comparison, and a follow-up edit run (20 further facts) uses the traced layer, the middle layer (L//2, with L the number of layers; layer 14 in Llama, 13 in Gemma) and a random layer.

Extra relations and rewording. Two relations whose answers differ across languages (landmark_country, city_country) add 34 facts for Llama and one for Gemma (A9). Phrasings 1 and 2 are traced for the first 30 valid facts of each model to test robustness to wording (H8).

4 Results

4.1 Overlap against the null

p-value floor. With 1000 permutations the smallest possible p is 0.001, so "p = 0.001" means "p $\leq$ 0.001"; with 10,000 permutations (Section 6, A14) the EN-ES p is 0.0001 in both models, the new floor.

model	pair	MLP cosine [95% CI]	null mean	z pooled	z within relation
Llama ($n = 92$)	EN-ES	0.627 [0.557, 0.702]	0.519	4.22	3.29
	EN-HI	0.509 [0.335, 0.594]	0.429	2.45	1.79
	ES-HI	0.423 [0.313, 0.529]	0.328	2.47	1.98
Gemma ($n = 58$)	EN-ES	0.464 [0.329, 0.542]	0.292	4.81	3.85
	EN-HI	0.226 [0.168, 0.283]	0.136	2.19	1.68
	ES-HI	0.137 [0.014, 0.267]	0.134	0.08	−0.45

Table 1: Pre-registered overlap analysis: median MLP-profile cosine across facts, shuffled-fact null mean, and z against the pooled and the within-relation null.

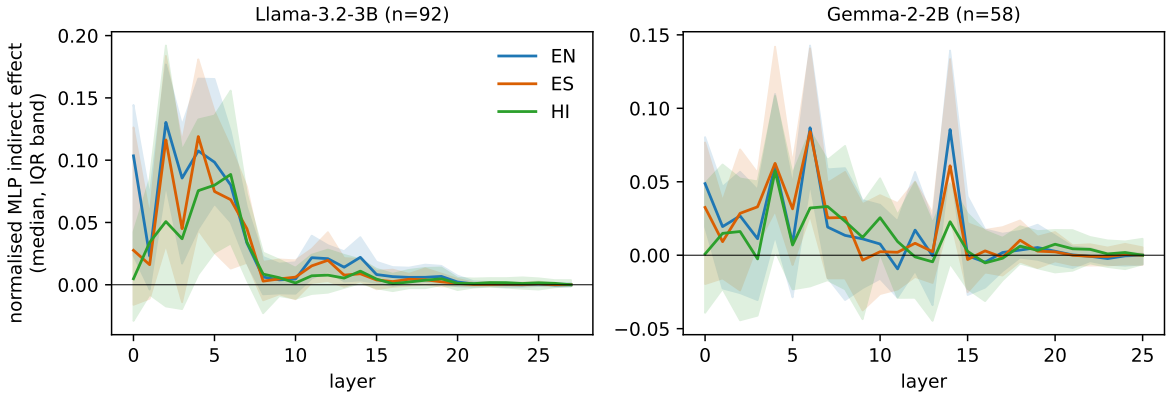


Figure 1: Median normalised MLP indirect effect per layer at each language's last subject token (band: interquartile range), for Llama-3.2-3B (left) and Gemma-2-2B (right).

Table~1 gives the pre-registered overlap results and Figures~1 and~2 show the profiles and the overlap. English-Spanish overlap is the only pair that survives the relation-stratified null and correction in both models. Within relation and after correction, only EN-ES remains significant in both models. Spearman statistics agree in direction (Llama EN-ES z = 5.15; Gemma EN-ES z = 4.55). Per relation with its own null the samples are small (6-38 facts) and mostly not significant (Llama capital_country EN-ES z = 2.75, capital_of 2.08, event_year 0.91). On the 33 facts both models keep, EN-ES is above the null in both (Llama z = 1.88, Gemma z = 4.09) and EN-HI is at the null in both (z = −0.21, 0.55).

4.2 Where the fact is looked up

The critical MLP layer (argmax at the last subject token) has median 3 (EN), 4 (ES), 4 (HI) in Llama (28 layers; 2-11% of facts in layers 8-19) and 6, 6, 6 in Gemma (26 layers; 26-43% in layers 8-19). The median absolute layer difference between languages is 2 in Llama and 2.5-4 in Gemma. The pre-registered expectation of a peak in layers 8-19 is not met for Llama. The median hides structure in Gemma (Figure~3, exploratory): the critical layer has modes at layers 4, 6 and 14; layer 14 holds the argmax for 17 of 58 English and 13 of 58 Spanish facts but for only 2 of 58 Hindi facts, and the cluster appears in three relations (English counts 3, 5, 9 in capital_country, capital_of, event_year). Llama has no such late cluster (3%, 2%, 11% of facts in layers 8-19 for EN, ES, HI).

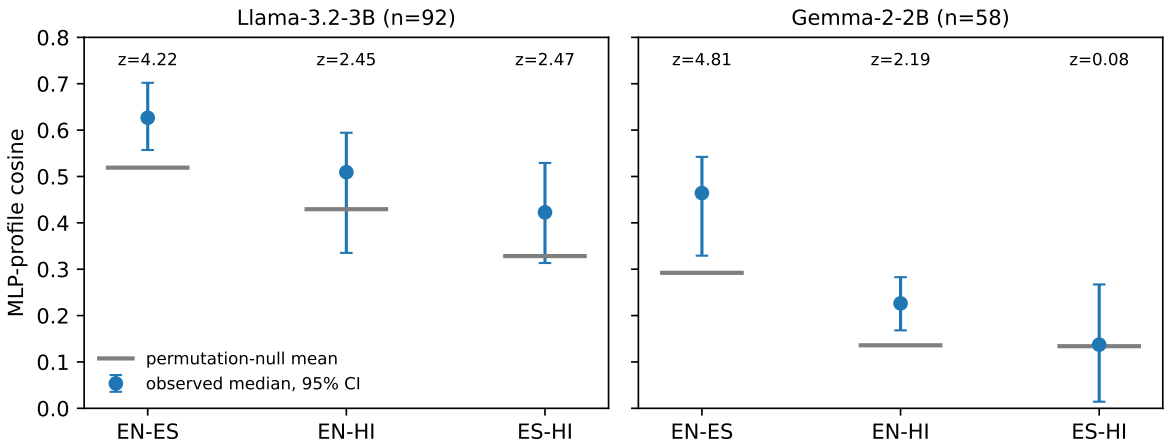


Figure 2: MLP-profile cosine between languages: observed median with 95% bootstrap CI (dots) and mean of the shuffled-fact permutation null (bars).

model	set	n	EN-ES	EN-HI	ES-HI
Llama-3.2-3B	all valid facts	126	0.642 (5.06)	0.500 (2.90)	0.413 (2.63)
Llama-3.2-3B	non-copyable	104	0.653 (4.87)	0.529 (2.78)	0.488 (3.51)
Llama-3.2-3B	all valid, stratified null	126	0.642 (4.24)	0.500 (2.66)	0.413 (1.98)
Gemma-2-2B	all valid facts	59	0.436 (3.86)	0.222 (2.09)	0.132 (0.06)
Gemma-2-2B	non-copyable	29	0.507 (2.53)	0.314 (1.16)	0.280 (0.42)
Gemma-2-2B	all valid, stratified null	59	0.436 (3.18)	0.222 (1.69)	0.132 (-0.40)

Table 2: Extended overlap (with the added relations). Median MLP-profile cosine (z against the shuffled-fact null). Stratified: permutations within relation. Gemma gained one fact, so its non-copyable set ($n = 29$) is below the pre-registered 50.

4.3 Hindi tokenisation

Hindi subjects span a median of 5 (Llama) and 8 (Gemma) tokens against 2 for English. The EN-HI profile cosine correlates negatively with Hindi subject span length (Spearman -0.35 Llama, -0.39 Gemma). Splitting at the median span, Llama's EN-HI cosine is 0.597 for short spans (z = 1.69 pooled) and 0.199 for long spans (z = -1.22). So part of the weak Hindi result is a span-length effect; it does not explain it away, because EN-HI stays below EN-ES with short spans in both models. Hindi accuracy drops mainly for answers of 7 or more tokens (Figure~4; Llama 0.23 against 0.40-0.42; Gemma 0.02 against 0.21).

4.4 Edit transfer over layers

For 10 facts per model (stratified by relation, English prompt only) we fitted a rank-1 edit of the MLP down-projection at every layer (280 edits in Llama, 260 in Gemma), toward a new object of the same relation, and scored greedy generation of the full new answer (Figures~5 and~6). Per-layer rates rest on 10 facts and move in steps of 0.1.

English. The edit makes the model produce the new answer for 84% of Llama edits and 74% of Gemma edits; it often fails at the last layers (final three layers: 53% success in Llama, 20% in Gemma). *Transfer.* Averaged over all layers, the new answer appears in Spanish for 41% (Llama) and 48% (Gemma) of edits and in Hindi for 18% and 20%. Per layer, Hindi success never exceeds 0.5 (Llama layer 4; Gemma layers 12, 13, 17). *Conditions (A5).* Table~3 lists the three conditions. At the fact's English traced layer, English

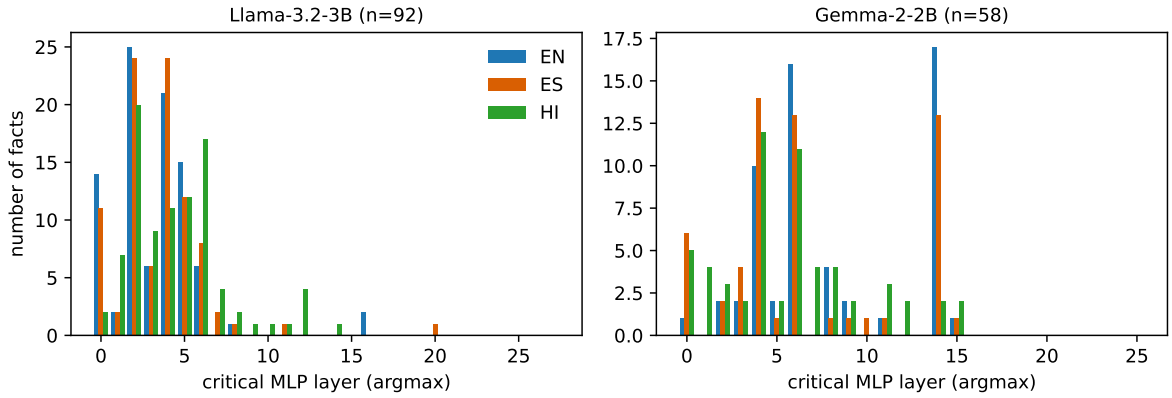


Figure 3: Distribution of the critical MLP layer (argmax of the profile) per language. Exploratory: Gemma shows modes at layers 4, 6 and 14.

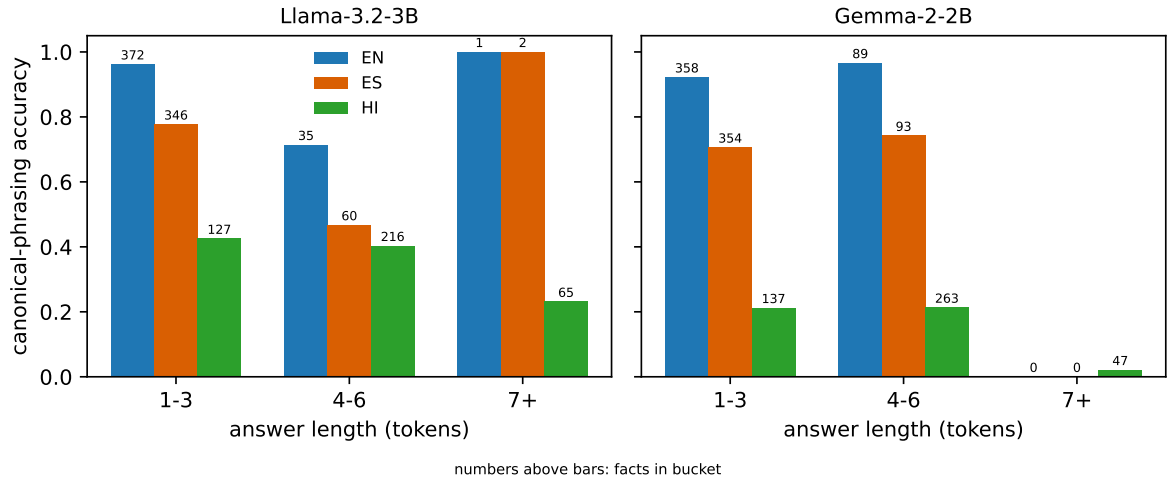


Figure 4: Canonical-phrasing accuracy by answer length in tokens, per language and model.

success is 0.9 in both models; median Spanish transfer ratio (dP_ES / dP_EN, over successful edits) is 0.66 in Llama and 0.38 in Gemma, and Hindi 0.0001 and 0.095. At L//2 (14 in Llama, 13 in Gemma) English success is 1.0 and 0.8, Spanish transfer 0.52 and 0.50, Hindi 0.010 and 0.016; specificity loss (maximum absolute probability change over 20 unrelated facts) is 0.084 and 0.089 against 0.030 and 0.027 at the traced layer. Median perplexity on 100 held-out sentences changes by at most 0.2% (Llama) and 0.6% (Gemma) at any layer. *H7 (mid-layer edits transfer more than layers 0-5, and cost more specificity).* The result depends on the model. In Gemma, near-mid layers (L//2 $\pm$ 3) transfer more than layers 0-5 to Spanish (median paired difference +0.34, 95% CI [0.01, 0.62]) and to Hindi (+0.07 [0.03, 0.13]) and lose more specificity (+0.04 [0.008, 0.19]); 8 of 10 and 9 of 10 facts have a positive Spanish and Hindi difference. In Llama the Spanish difference is negative (-0.31 [-0.47, -0.09]; 1 of 10 facts positive), the Hindi difference is negative or zero (-0.013 [-0.095, 0.000]) and the specificity difference includes zero (+0.004 [-0.023, 0.035]). H7 is therefore supported in Gemma and contradicted in Llama; we make no pooled claim.

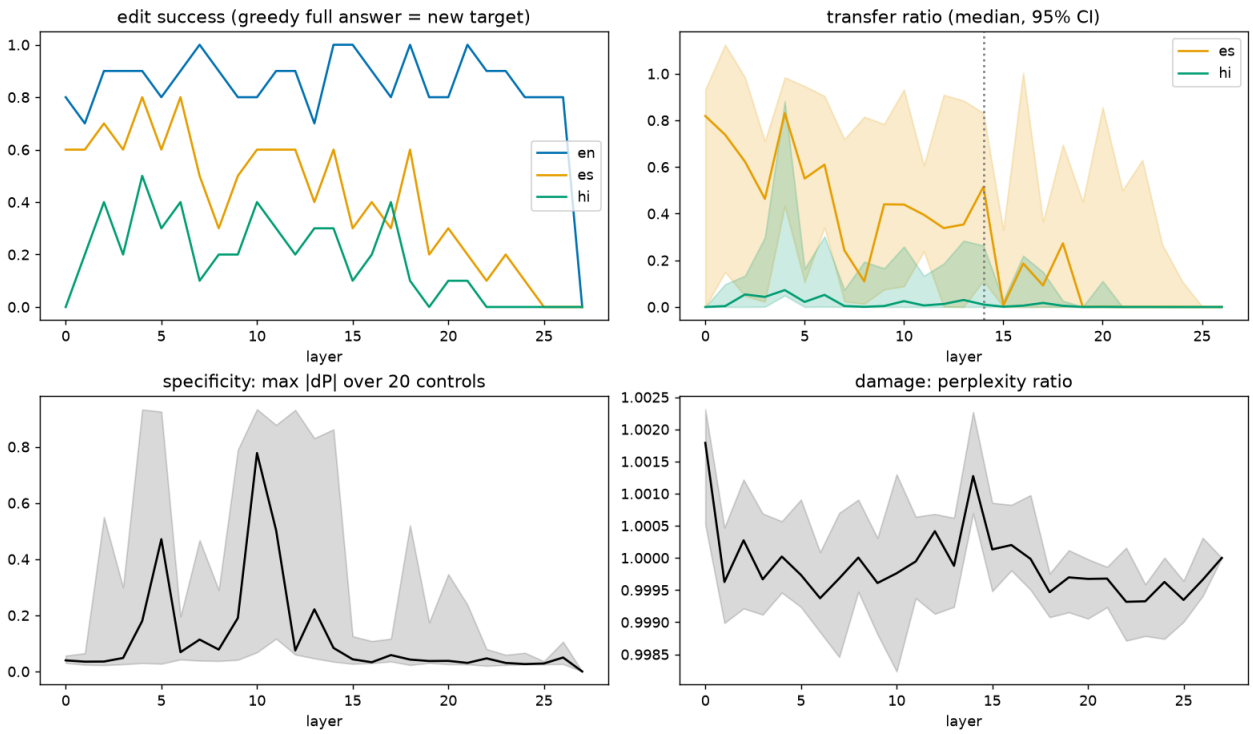


Figure 5: Llama-3.2-3B edit sweep (10 facts, every layer): English success, transfer ratio, specificity and perplexity ratio by layer.

4.5 Robustness: phrasings (H8), non-copyable relations (F4), in-context baseline (F3), follow-up edits (F7)

Phrasings (H8). For the first 30 valid facts of each model we traced two further phrasings in all three languages. Within a language, the MLP-profile cosine between phrasings of the same fact exceeds the cross-language cosine of the canonical phrasings: Llama 0.754 against 0.609 (median per-fact difference 0.139, 95% CI [0.101, 0.228]; 26 of 30 facts positive), Gemma 0.741 against 0.338 (0.341 [0.287, 0.463]; 30 of 30). H8 is supported in both models. By language the within-language median is 0.985 (EN), 0.744 (ES), 0.695 (HI) in Llama and 0.819, 0.778, 0.713 in Gemma. The profile therefore reflects the fact and language more than the wording, and cross-language overlap is lower than the change of wording alone.

Non-copyable relations (F4). We added landmark_country and city_country, whose answers differ in form across languages. Llama keeps 34 extra facts (126 valid in total; 104 without event_year and invention_inventor, so the target of at least 50 is met). On the 104 non-copyable facts (Table~2) the MLP-profile cosine is 0.653 (EN-ES, z = 4.87), 0.529 (EN-HI, z = 2.78) and 0.488 (ES-HI, z = 3.51) against a shuffled-fact null (pooled within this set). With the relation-stratified null on all 126 facts the z-values are 4.24, 2.67 and 1.98. Within single relations in Llama, EN-ES z is 2.75 (capital_country, n = 38), 2.08 (capital_of, 26), 1.96 (city_country, 27) and 0.91 (event_year, 22). Gemma keeps only one extra fact, so its non-copyable set has 29 facts and the target is not reached; there EN-ES is 0.507 (z = 2.53), EN-HI 0.314 (z = 1.16) and ES-HI 0.280 (z = 0.42). The English-Spanish alignment therefore does not depend on copyable answers in Llama; for Gemma the non-copyable sample is too small for a conclusion.

In-context baseline (F3). On 10 facts per model we placed the new fact in an English statement before the question (descriptive, no hypothesis). In Gemma the English statement changed the answer in English for 8/10 facts, Spanish 7/10 and Hindi 6/10 (median probability change 0.48, 0.56, 0.27). In Llama it changed Spanish and Hindi answers for 10/10 facts each (0.75, 0.69) but English for 0/10 (0.003); a diagnostic on the first 6 facts

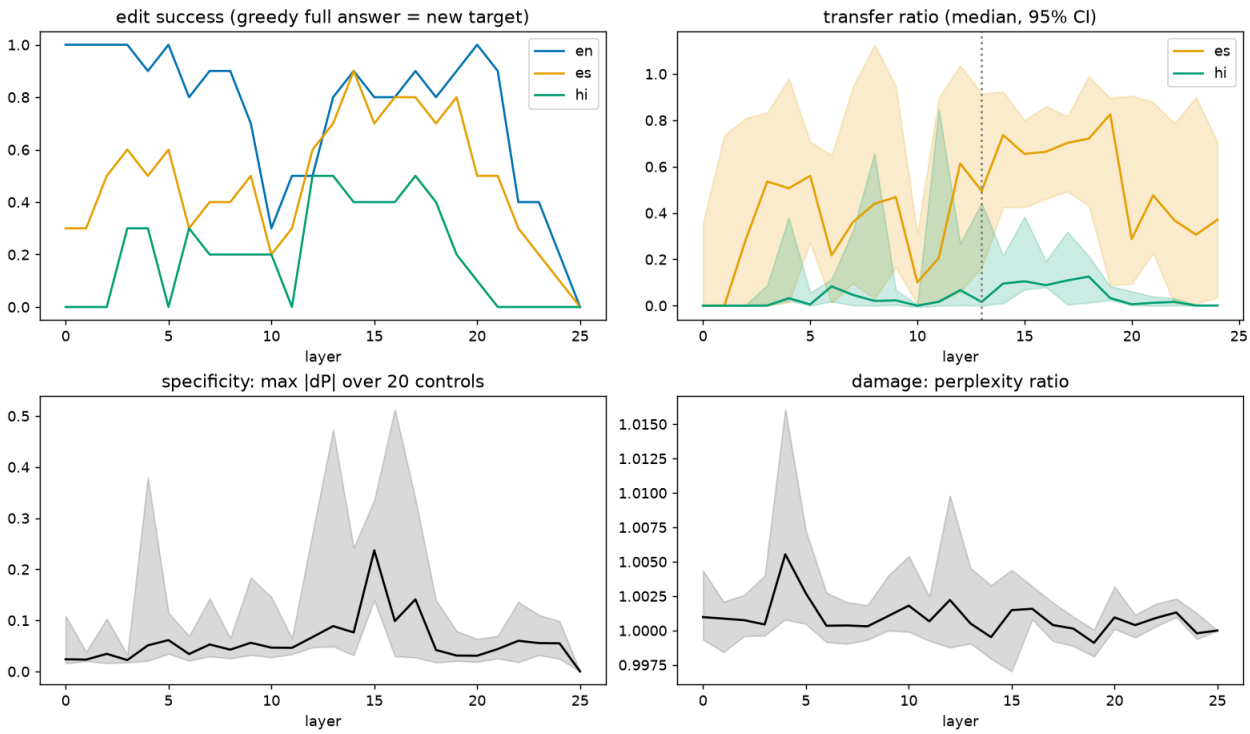


Figure 6: Gemma-2-2B edit sweep, same layout as Figure 5.

(greedy generations saved in `analysis/run_20261025_incontext_diag_llama/generations.txt`) shows why: in English the model writes the original answer and rejects the new one (for example "France, not Denmark"), whereas in Spanish it repeats the new object. The English test prompt repeats the context sentence word for word, which invites a correction; the 0/10 is therefore a behaviour of this prompt design in an instruct model, not a code error, and the English figure should not be read as the model being unable to use the context. In Llama this prompt-level transfer to Spanish and Hindi exceeds that of the weight edit (41% and 18% of edits on average over layers, section 4.4).

More edit facts (A20). Twenty further Llama facts were edited under the same three conditions (traced layer, layer 14, layer 21) and pooled with the F7 facts (40 facts; the original 10 sweep facts are not pooled). English success is 0.80 (traced), 0.93 (layer 14) and 0.72 (layer 21). Median Spanish transfer is 0.58 [0.11, 0.88], 0.75 [0.64, 0.91] and 0.00; Hindi transfer is 0.00, 0.10 [0.01, 0.36] and 0.00. Median specificity loss is 0.038, 0.148 and 0.029; the perplexity ratio stays within 0.9996-1.0004. The layer-14 edit therefore transfers more to Spanish and succeeds more often, at about four times the specificity loss; the registered prediction that traced and layer-14 transfer would be within 0.15 was not met. Descriptive, one model, instruct only; the traced-layer CI is wide (transfer is bimodal across facts).

Follow-up edits (F7). For 20 further Llama facts (30 with the sweep facts) edited at the fact's traced layer, L//2 and the A9-selected layer (layer 21, labelled "random" in the run files because it is drawn from the swept layers other than the traced and L//2 ones, and it was layer 21 for all 20 facts), English success is 0.80, 0.95 and 0.65. Median Spanish transfer ratio is 0.70 (traced) and 0.78 (L//2), Hindi 0.005 and 0.105. Median specificity loss is 0.062 at the traced layer and 0.568 at L//2 (95% CI [0.08, 0.78]); median perplexity changes by less than 0.05% at the traced layer and L//2. Spanish transfer at layer 21 is near zero (median ratio 6.6e-5). The same follow-up for Gemma (20 facts per condition; 19 at layer 3) gives English success 0.80 (traced), 0.65 (L//2) and 0.89 (layer 3). Median Spanish transfer ratio is 0.65, 0.42 and 0.66; Hindi 0.10, 0.20 and 0.0005. Median specificity loss is 0.084, 0.078 and 0.032; median perplexity changes by less than 0.06% in

model	layer	n	EN success	ES transfer	HI transfer	specificity
Llama-3.2-3B	traced	10	0.90	0.664	0.000107	0.0303
Llama-3.2-3B	mid	10	1.00	0.515	0.0104	0.0838
Llama-3.2-3B	random	9	0.78	0.419	0.00658	0.0587
Gemma-2-2B	traced	10	0.90	0.376	0.0954	0.0265
Gemma-2-2B	mid	10	0.80	0.495	0.0161	0.0888
Gemma-2-2B	random	6	0.67	0.0933	0.0364	0.129
Llama-3.2-3B, follow-up	traced	20	0.80	0.696	0.00507	0.0624
Llama-3.2-3B, follow-up	mid	20	0.95	0.78	0.105	0.568
Llama-3.2-3B, follow-up	A9 layer	20	0.65	6.58e-05	2.5e-05	0.0281
Gemma-2-2B, follow-up	traced	20	0.80	0.652	0.0998	0.0843
Gemma-2-2B, follow-up	mid	20	0.65	0.423	0.196	0.0778
Gemma-2-2B, follow-up	A9 layer	19	0.89	0.655	0.000512	0.032

Table 3: Edit conditions: the fact's English traced layer, $L/2$ and a third layer (medians; in the 10-fact sweeps this is a randomly drawn layer, in the follow-ups it is the layer chosen by the A9 rule: 21 for Llama, 3 for Gemma; transfer = $\Delta P_{L2}/\Delta P_{EN}$; specificity = maximum absolute probability change over 20 unrelated facts). The first two blocks are the 10-fact sweeps; the last two are the 19-20-fact follow-ups.

all three. Unlike Llama layer 21, the Gemma A9-selected layer (layer 3) edit transfers to Spanish. In this follow-up the "random" condition is drawn from the swept layers other than the traced and L//2 ones; for Gemma those are the A9-selected layer 3 and, for some facts, a few other layers, and it was layer 3 for 19 of the 20 facts, so this condition is effectively the A9-selected layer, not an independent random layer. These are descriptive medians on 19-20 facts. H7 is not re-tested on them.

4.6 Controls added after review (A11)

An external review argued that the English-Spanish similarity could reflect generic profile shape or subject-name overlap. We pre-registered controls (A11, before computing) and ran them on the cached traces (Llama 126 facts, Gemma 59; 10,000 permutations; no new model run).

Same-language baselines. In Gemma the same-fact EN-ES cosine is 0.424 against 0.242 for different facts of the same relation (EN-EN 0.247, ES-ES 0.234). In Llama it is 0.570 against 0.480 for different-fact EN-ES, but 0.606 for EN-EN and 0.413 for ES-ES: English profiles of different facts are more alike than English and Spanish profiles of the same fact, so in Llama the profile shape is strongly language-specific and the fact-specific increment (0.09) is a small part of the cosine.

Centred profiles. After subtracting each language's mean profile, the EN-ES median cosine is 0.284 (null 0.113, z = 3.61) in Llama and 0.410 (null 0.082, z = 5.43) in Gemma. The alignment is not an artefact of the shared mean profile.

Lexical overlap. After removing accents and case, 36.5% of Llama subjects and 11.9% of Gemma subjects are identical in English and Spanish. The per-fact EN-ES cosine correlates with the similarity of the two subject strings in Llama (Spearman 0.29, permutation $p < 0.001$) and weakly in Gemma (0.22, $p = 0.097$). Split at the median similarity, Llama's EN-ES alignment is z = 4.50 (uncentred) and 3.36 (centred) in the high-similarity half (n = 62) but 2.04 and 1.44 in the low-similarity half (n = 64); the centred value is below the pre-registered threshold of 2. Gemma's low-similarity half gives z = 2.80 and 3.42 (n = 30). On these single-seed profiles, a substantial part of Llama's EN-ES similarity is therefore associated with subject-name overlap, and the remainder is not distinguishable from zero after centring at this sample size; in Gemma it is. The independent-noise re-analysis below revises this for Llama.

Reconciling this split with the later covariate model (A22). The split above used the original single-seed

profiles, which Section 4.7 shows to be noisier (A13a: mean increment 0.239 against 0.346 with independent noise, Llama). We therefore repeated the split on the independent-noise profiles (registered as A22 before the computation; same facts, no new model run). Splitting the 103 informative Llama facts by whether the English and Spanish subject spans share any subword piece (Jaccard 0 against above 0), the fact-specific increment is present in both halves: 0.303 [0.192, 0.423] (n = 54, no shared piece) and 0.397 [0.291, 0.505] (n = 49, shared piece); leak-free 0.312 [0.202, 0.441] (n = 53). In Gemma the no-shared-piece half has 0.264 [0.139, 0.402] (n = 21) and the shared-piece half has only 4 facts. So in Llama subject-name overlap raises the increment by about 0.09 but is not required for it; the sentence above that the remainder "is not distinguishable from zero" applies to the single-seed profiles and is superseded by this result. The pre-registered prediction that Gemma's halves would be mostly unresolved was not met (the no-shared-piece half is present). Of the four pre-registered predictions, the two about Llama held, the one about Gemma did not, and the leak-free prediction held for Llama but not for the small Gemma halves (one changed from present to unresolved, one has no facts left).

Entity dependence. Clustering facts that share an English subject or answer (81 clusters in Llama, 56 in Gemma) leaves the 95% CI of the median EN-ES cosine essentially unchanged (Llama [0.596, 0.688]; Gemma [0.323, 0.536]).

4.7 Is the alignment shared retrieval? Noise re-trace and interchange test (A12-A13)

Shared-noise check (A13a). The original tracer used one noise seed per fact for every language, which could inflate EN-ES similarity. We re-traced both models with independent noise per language (10 draws each; the last-token MLP profile, same score). The fact-specific EN-ES increment D (same-fact cosine minus the mean for other facts of the same relation) stays clearly above zero: Llama 0.346 (97.5% CI [0.271, 0.424], n = 103 informative facts; 0.367 [0.293, 0.445] on all 126), Gemma 0.257 ([0.130, 0.381], n = 25; 0.297 [0.214, 0.374] on all 59). With the original single seed the same facts gave 0.239 (Llama) and 0.128 (Gemma), so the single-draw profiles were noisier, not inflated. Within-language noise ceilings (draws 0-4 against 5-9) are 0.82/0.79 (Llama EN/ES) and 0.59/0.54 (Gemma), so most of the remaining gap is language difference, not noise.

Covariate control (A13b). Adding subject-token overlap, answer identity, span length and effect size as covariates leaves Llama's increment at 0.319 ([0.210, 0.428], n = 103). Gemma is unresolved at n = 25 informative facts (0.138, [−0.086, 0.362]) and present on the 55 non-identical-prefix facts (0.216, [0.047, 0.385]).

Interchange (A13). Profile similarity does not show that the same representation is read out. We therefore swapped the residual stream at the last subject token (site A) or final token (site B), one layer at a time, from a source prompt in one language into a target prompt in the other, and measured the change in log-probability of the source's answer minus the target's answer. The statistic I is the cross-language effect divided by the same-language effect, summed over the layers where the same-language effect is at least half its maximum. Confirmatory pairs have source subjects with no shared token between English and Spanish (Llama 142 pair-directions, 67 source entities; Gemma 162, 71); CIs resample source entities. The same-language gate (flip rate at least 60%, effect at least 2 nats) is met in all eight cells (flip 0.68-0.95, effect 19-27 nats).

At site A the ratio is Llama EN-to-ES 1.17 [1.09, 1.28] and ES-to-EN 0.90 [0.86, 0.94]; Gemma 1.07 [1.03, 1.12] and 0.94 [0.91, 0.97]. All four lower bounds exceed the pre-registered 0.7, so the pre-registered rule gives SHARED. The answer flips in 85% and 67% of Llama cases and 90% and 87% of Gemma cases; swaps from the same fact (self) and from a wrong token (control) flip none. At site B the ratio is also near 1 (0.90-1.11) but the cross-language flip rate is only 34-45%, against 67-94% within a language, so the final-token state is less interchangeable than the subject-token state. The lexical-overlap prediction is not met: the fitted ratio falls from 1.17 to 0.80 (Llama) and 1.03 to 0.91 (Gemma) from zero to full token overlap.

Caveats. Four relations (capital of country, capital country, city country, landmark city), English and

Spanish only, and the identity check differs from zero by up to 0.119 nats (bf16 batch numerics). The tracing increments above do not predict interchange success per fact: the Spearman correlation is -0.04 in Llama (n = 40, p = 0.83), so the pre-registered metric-validity check fails there; in Gemma it is 0.23 on only 9 facts. Tracing similarity and interchange therefore support the same conclusion for these relations without being shown to measure the same quantity.

Hindi interchange (A15, underpowered). We repeated the test with Hindi in place of Spanish. Hindi recall limited the fact set to 30 Llama facts (all capital-of) and 16 Gemma facts (all capital-country), giving 23 and 12 source entities, below the pre-registered floor of 25, so we make no claim. In the pre-registered primary direction (English source into a Hindi prompt) the same-language ceiling is too weak (answer flips in 12% of Llama and 36% of Gemma cases against the required 60%), so the gate fails in both models. In the reverse direction the gate passes (flip 0.80 and 0.82) and the site-A ratio is 0.74 [0.67, 0.81] in Llama and 0.74 [0.69, 0.82] in Gemma, with cross-language flip rates of 0.26 and 0.45; by the rule this is PARTIAL, but we report it as descriptive only. The Hindi question therefore stays open.

Hindi interchange, re-test (A16). Part of the A15 failure was format: the Hindi templates made Llama continue with a question mark and a new line, which the strict answer match counted as wrong. We added question-and-answer templates, dropped the per-relation threshold, tested only Hindi-to-English (the direction whose ceiling passed) and used four source facts per target. Llama kept 66 facts (three relations) and Gemma 55 (four), giving 66 and 54 source entities, above the floor of 25. At site A the ratio is 0.74 [0.69, 0.78] in Llama (answer flips 0.48 against 0.80 within Hindi; PARTIAL, since the lower bound is below 0.7) and 0.76 [0.71, 0.81] in Gemma (flips 0.64 against 0.90; SHARED, by a margin of 0.01). On facts that were not in the A15 set the ratios are 0.76 and 0.77. Per relation the ratio ranges from 0.53 (Gemma, landmark city) to 0.87 (Gemma, city country). At site B the cross-language flip rate is 0.00, so the ratio is not interpretable. Hindi-to-English retrieval is therefore largely but not fully shared, weaker than English-Spanish (0.90-0.94), and the two models fall on either side of the 0.7 threshold with the same point estimate; English-to-Hindi was not tested here because patching one token cannot carry a 5-8 token Hindi entity; the span-wise test below does.

Hindi interchange, span-wise (A18). We patched the whole subject span: the last k subject positions of the target prompt (k = the shorter of the two subject lengths) receive the source's last k subject-position states, end-aligned, in both directions on the A16 fact sets (200 pairs per model and direction; 190 confirmatory pairs and 60 source entities in Llama, 186 and 53 in Gemma). The same-language ceiling now passes in all four cells (answer flips 0.71 EN to HI and 0.94 HI to EN in Llama; 0.81 and 0.95 in Gemma). The site-A ratio is 0.86 [0.80, 0.93] (EN to HI) and 0.87 [0.82, 0.92] (HI to EN) in Llama, and 0.93 [0.88, 0.99] and 0.81 [0.76, 0.86] in Gemma; all four are SHARED by the pre-registered rule and the two directions differ by 0.01 and 0.13, within the registered symmetry margin of 0.15. Excluding facts with the answer inside the subject changes the ratios by at most 0.05 (largest: Gemma HI to EN 0.81 to 0.85, EN to HI 0.93 to 0.95). The label is the same under all four bootstrap clusterings (source, target, connected component, two-way) in all eight cells (plain and leak-free); the smallest lower bound is 0.703 (Gemma HI to EN, component clustering), so that cell is close to the threshold. Patching the span moves the Hindi-to-English estimate up from 0.74-0.76 (last token only), so the earlier number understated the shared fraction. *Base-model replication (A19).* To test whether the result depends on instruction tuning we repeated the English-Spanish and English-Hindi interchange on Llama-3.2-3B base (same architecture and tokenizer; 81 English-Spanish facts and 76 Hindi facts kept, 200-pair cap, 79 and 190 confirmatory pairs, 33 and 69 source entities). The same-language ceiling passes in all four directions (flip 0.66-0.91). Site-A ratios: English to Spanish 1.21 [1.11, 1.37], Spanish to English 0.87 [0.81, 0.92], Hindi to English 0.86 [0.82, 0.90] (all SHARED) and English to Hindi 0.72 [0.68, 0.77] (PARTIAL: lower bound below 0.7). They differ from the instruct model's by 0.04, 0.03, 0.01 and 0.14, all within the registered margin of 0.15, so all four cells replicate; English to Hindi is the weakest and the one nearest the margin. Labels are the same under all four bootstrap clusterings and with answer-leaking facts removed (changes ≤ 0.01). The Spanish estimates rest on fewer facts than the instruct run (the 100-fact

model	direction	pairs	l^*	flip same	flip cross	I [95% CI]	verdict
Llama-3.2-3B	EN→ES	142	6	0.68	0.85	1.17 [1.09, 1.28]	SHARED
Llama-3.2-3B	ES→EN	142	5	0.84	0.67	0.90 [0.86, 0.94]	SHARED
Gemma-2-2B	EN→ES	162	6	0.87	0.90	1.07 [1.03, 1.12]	SHARED
Gemma-2-2B	ES→EN	162	6	0.95	0.87	0.94 [0.91, 0.97]	SHARED

Table 4: Interchange test (site A, confirmatory pairs whose source subject shares no token piece across languages). I = cross-language effect divided by same-language effect, summed over the layers where the same-language effect is at least half its maximum; CI by bootstrap over source entities. Verdict by the pre-registered rule (lower bound ≥ 0.7: shared).

prediction was missed).

The broadcast variant (last-token state copied over every position) fails the ceiling in Llama (flip 0.41) and is not interpreted. These ratios are estimates from 3-4 relations and two models, not a general rate.

5 Discussion

What the evidence supports. In both models the English and Spanish causal profiles of the same fact are aligned beyond a shuffled-fact null, and the alignment survives a relation-stratified null, rewording of the question (H8), and, in Llama, restriction to answers that are not copyable across languages. This is consistent with partly shared lookup for these two languages, and the lexical control (Section 4.6) shows that in Llama the alignment is stronger for facts whose English and Spanish subject names share subword pieces (0.40 against 0.30) but still present for those that do not. Hindi is weaker: its overlap with English is above the pooled null in Llama ($z = 2.90$ on 126 facts) but not significant after the stratified null with Bonferroni correction ($z = 2.67$, $p = 0.012$), and depends on how many tokens the Hindi subject takes.

Interchange (Section 4.7). Swapping the subject-token state between English and Spanish prompts moves the answer nearly as much as a same-language swap (ratio 0.90-1.17 in both models; pre-registered verdict SHARED for English-Spanish on four relations), which supports shared retrieval of the fact at the subject token beyond profile similarity. Patching the whole Hindi subject span gives ratios of 0.81-0.93 in both directions and both instruct models, and the Llama base model reproduces all four cells within 0.15 (English to Hindi, 0.72, is the weakest and PARTIAL), so the result does not depend on instruction tuning for Llama. Two readings agree with earlier work that script matters (Ifergan et al., 2024; Beniwal et al., 2024): the single-token Hindi estimate was lower, and Hindi edit transfer is near zero. The per-fact profile increments do not predict per-fact interchange success in Llama, so the two measurements agree in direction, not in detail.

What it does not support. Profile similarity at one token and one component is not identity of mechanism, and we found no evidence of a single language-agnostic space. Edits made in English reach Spanish for about half of the edits and Hindi rarely; transfer depends on the layer and on the model (H7 supported in Gemma, contradicted for Spanish in Llama). The in-context baseline shows that the model can use an English fact in Spanish and Hindi when it is in the prompt (Llama), so the low Hindi transfer of weight edits is not simply a lack of cross-lingual access; the edit site and the way Hindi is tokenised are alternative explanations that we have not separated.

Shared at the subject token, yet edits carry over only partly. The interchange ratio for Spanish and Hindi is near 1, but the weight edit carries over to Spanish only in about half of the cases and to Hindi rarely. Taken together, these results say that a shared subject representation is not sufficient for a rank-1 edit of one language to reach another; where the edit is written matters (layer 14 against the traced layer; Hase et al., 2023). The paraphrase control (A21) that would show whether the Spanish shortfall exceeds the shortfall to new English wordings has not yet been run.

Cost of mid-layer edits. Pooled over 40 Llama facts (Section 4.5, A20), edits at layer 14 have higher median Spanish transfer than edits at the traced layer (0.75 [0.64, 0.91] against 0.58 [0.11, 0.88]; the intervals overlap) and succeed more often in English (0.93 against 0.80), but have a higher median specificity loss (0.148 against 0.038); any transfer gain is therefore not free.

Scope. Two model families (plus one base-model replicate), hand-written facts, interchange on four relations (English-Spanish) and 3-4 relations (English-Hindi, both directions, span-wise), 10 to 40 edit facts per condition and a small Gemma non-copyable sample limit generality. The BLOOM-1b7 and Llama-3.2-1B pilots (A10), which test whether the weak Hindi result is a tokenisation effect, had not run when this draft was written.

6 Deviations from the pre-registration

H6 (EN-ES z > 2 and EN-ES > EN-HI) holds in both models. H8 (phrasing robustness) is supported in both models. H9 (EN-ES z > 2 in at least two new or extended relations) is not supported: only event_year was testable among the pre-registered relations (z = 0.91); the later city_country relation reaches z = 1.96, just below the threshold. The F4 target of 50 non-copyable facts is met for Llama (104) and not for Gemma (29). The F7 follow-up (30 edit facts) finished for Llama only (20 new facts); the Gemma run was interrupted and its partial output is kept unused. The expected layer 8-19 peak is not met for Llama. Amendments A1-A10 (docs/HYPOTHESIS.md; A10, the BLOOM-1b7 pilot, is pre-registered but not yet run): critical-site definition; competitor from the corrupted run (v1 only); statistics definition; edit design; v2 design; matching rule; template selection after a poor first filter; Gemma pool extension; analysis rules (stratified null as primary for new data, non-copyable set, in-context baseline). Later amendments A11-A22 are in `docs/HYPOTHESIS.md`, each registered before its data or computation; A22 (the split by name overlap on independent-noise profiles) was run on stored data after A11 had shown the first split, which is disclosed in Section 4.6. The stratified null and the Hindi span split were added after seeing results and are labelled post hoc. v1 first-token results (66 facts, Llama only) are kept in `paper/paper.md` as a pilot; in v1 the Hindi pairs were at the null, which the v2 analysis shows to be sample- and metric-dependent.

Permutation count (A14). The pre-registered overlap tests use 1000 permutations, so the smallest possible p is 0.001. We reran the same traces with the same seed and 10,000 permutations (CPU, no new model run). Observed medians are identical. For the MLP-profile cosine, z changes by at most 0.21 on the main-table sets (EN-ES: Llama 4.22 to 4.01, Gemma 4.81 to 4.82; p = 0.0001 in both) and no verdict changes; the stratified Gemma EN-HI p is 0.0502 (0.050 before), still not significant. The top-5 IoU z is unstable across draw counts (Llama 14.9 to 10.9 on the 126-fact set), so we do not rely on it. Two secondary tests at p of about 0.05 cross 0.05 (Gemma no-copyable EN-ES Spearman 0.051 to 0.043; Gemma event_year EN-HI 0.051 to 0.050). The tables in this paper keep the 1000-draw values; the 10,000-draw values are in `results/manifest.json` (keys `pk_*`).

7 Limitations

Two models, one of them a base and one an instruct model; hand-written facts; small per-relation samples; event_year answers are digit strings that can be copied across languages and make up half of Gemma's set; Hindi is hard to trace because subjects span many tokens; invention_inventor keeps 0 facts in both models. Layer-0 edit results were not reproducible between runs (v1). The EN/ES/HI facts were written by the authors' automated assistant and have not been checked by a native speaker; this is the main validity limitation of the data. Five relations, mostly geography; three languages; models of 2-3B parameters; no 7B-class model. The edit comparison is descriptive (30-40 Llama facts, one model, instruct only) and has no paraphrase control yet. Multiple testing: with Bonferroni over the six primary overlap tests (alpha 0.0083) Gemma's

EN-HI ($p = 0.016$) is not significant. The tracing increments do not predict interchange success per fact in Llama (Spearman -0.04), so the two measurements are not shown to measure the same quantity. There is no external (OSF/Zenodo) pre-registration timestamp; the order of pre-registration and data is evidenced by git history only.

8 Reproducibility

Code, configs, write-once run directories, seeds, logs and a claim manifest (`results/manifest.json`, `./reproduce.sh verify`) are in the repository. End-to-end rerun status: see `docs/LOG.md`.

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